

DAY 3 · SESSION 5

Next-Generation Protection for Oracle and Enterprise Databases

Azure Storage, Data Protection, and Resilience at Scale

University of Minnesota

This morning, block by block

01 **Block 1 · 55 min**

Oracle on Azure deployment options

02 **Block 2 · 55 min**

Oracle backup to immutable Blob and the live demonstration

03 **Break · 15 min**

Coffee and a reset before the platform tour

04 **Block 3 · 40 min**

Other database platforms and restore validation

05 **Block 4 · 75 min**

Design exercise and leadership pitch review

The university's most critical databases deserve a modern protection strategy

The largest, most business-critical databases run on Oracle and support enterprise resource planning and identity management.

The current protection approach is decades old.

The team has been asked to define a modern strategy centered on immutability.



By the end of this session you can

Compare Oracle paths
Oracle Database@Azure and Oracle on Azure VMs, including licensing and performance.

Design Oracle backup architecture
Land RMAN backups in immutable Blob storage.

Describe protection options
SQL Server, Azure SQL, MySQL, and PostgreSQL.

Apply enterprise safeguards
Immutability, separation of duties, and restore testing for enterprise databases.

Build a business case leadership can act on.

Bringing Days 1 and 2 to the database tier



BLOCK 1 · 55 MIN

Oracle on Azure deployment options

Who runs the platform, which tools apply, and how licensing works.



Two main paths for Oracle on Azure

PATH A · ORACLE-MANAGED

Oracle Database@Azure

Now named Oracle AI Database@Azure. Oracle-managed database services on OCI infrastructure colocated in Azure datacenters.

PATH B · SELF-MANAGED

Oracle on Azure virtual machines

You install, operate, and protect Oracle Database on Azure IaaS.

The choice drives who operates the platform, which backup tools apply, and how licensing works.



Oracle AI Database@Azure: Oracle services inside Azure datacenters

- All service infrastructure sits in Azure's physical datacenters for low latency to Azure applications, in more than 30 Azure regions
- Database services: Exadata Database Service on Dedicated Infrastructure, Exadata Database Service on Exascale Infrastructure, Oracle Base Database Service, and Oracle Autonomous AI Database (serverless, or on dedicated Exadata infrastructure)
- Data and protection services alongside them: Oracle Database Zero Data Loss Autonomous Recovery Service, OCI GoldenGate, and Autonomous AI Lakehouse
- Database versions 19c and 26ai (23ai was renamed 26ai)
- Uses Azure virtual networks, Azure identity, and native Azure monitoring of metrics and logs
- Oracle's operations team handles patching and infrastructure updates through a connection to OCI

How Oracle Database@Azure is bought and managed

01

Purchase

Through Azure Marketplace as pay-as-you-go or a private offer.

02

Link tenancies

Link the Azure subscription to an OCI tenancy.

03

Manage

Infrastructure and VM clusters in the Azure portal; some container and pluggable database tasks in the OCI console.

04

Set boundaries

Clarify support boundaries between Microsoft and Oracle early.

Oracle on Azure VMs: full control, full responsibility

Compute and disks

Memory-optimized VM sizes paired with Premium SSD, Premium SSD v2, or Ultra Disk for data and redo.

File storage option

Azure NetApp Files as an NFS alternative for Oracle data files with high throughput.

What you own

Patching, high availability (for example Data Guard), and backups.

Why back up at all

All Azure storage is highly redundant, and redundancy copies mistakes too. Backups protect against human error, ransomware and insider deletion, support cloning, and meet regulatory needs.

Licensing and support boundaries

Licensing

- Oracle licensing on Azure VMs follows Oracle's authorized cloud environment policy; confirm with your Oracle account team
- Oracle Database@Azure purchase options include bring your own license and license-included, depending on service

Support

- Map who supports each layer: Azure infrastructure, OCI-managed services, database, and application
- Bring licensing into the business case, not after it

Choosing the Oracle path

Consideration	Oracle Database@Azure	Oracle on Azure VMs
Who runs the infrastructure	Oracle operations team	Your team
Platform	Exadata or Autonomous on OCI in Azure datacenters	Azure VMs with managed disks or Azure NetApp Files
Backup tooling	Oracle-managed backups, including the Zero Data Loss Autonomous Recovery Service; Azure targets per service options	RMAN, Azure Backup, storage snapshots
Best fit	Exadata-class performance with less operational burden	Maximum control and custom configurations

BLOCK 2 · 55 MIN

Oracle backup to immutable Blob and the live demonstration

From RMAN to a backup no one can alter, then a restore to prove it.



The Oracle backup toolbox on Azure



RMAN

Required to initiate Oracle backup and restore operations and copy backups to Azure Storage.



Azure Backup

Application-consistent VM backups of Oracle on Linux VMs.



Azure Files

Cost-effective RMAN target over SMB; Microsoft notes Azure Backup is better for large databases.



Blob Storage

RMAN target through a mounted file system, with immutability on the container.

Getting RMAN backups into Blob

RMAN writes to a file system path, so Blob must be presented as one.

1

BlobFuse2

Mounts a Blob container on Linux over the Blob REST API.

2

NFS 3.0 on Blob

Mounts Blob over NFS, but it requires a hierarchical namespace account, supports LRS and ZRS only, and immutability policies are not supported on NFS 3.0-enabled accounts.

3

Immutable design

Design the mount path around a flat-namespace container that supports WORM policies.

Making the backup container immutable

Time-based retention

Backup sets can be read but not modified or deleted for the interval.

Legal hold

For investigations and audits.

Version-level WORM

Requires blob versioning.

Align retention

Match RMAN retention and obsolete deletion, or deletes will fail.

Align RMAN retention with immutability

01

Set the policy

Set RMAN recovery window or redundancy policy.

02

Match WORM

Set the WORM retention interval equal to or longer than that window.

03

Delete safely

Let RMAN mark backups obsolete, and delete only after WORM retention expires.

04

Watch cost

Monitor capacity growth and cost per retained day.

Geo-protection for database backups



Another region

Without a DR technology like Data Guard, store backups in another region via geo-redundant replication.



Immutable destination

Combine geo-redundancy or object replication with immutability on the destination.



Separate subscription

Keep the backup account in a separate subscription from production.

Separation of duties for database protection

DBAS

Run RMAN.

STORAGE ADMINISTRATORS

Own the immutable target.

SECURITY

Owns the Resource Guard for vault-based backups and the lock decision for Blob WORM policies.

- No single role can both create and destroy the protected copy: DBAs write, storage administrators hold the container, security holds the lock.
- Use managed identities for RMAN mount authentication, never account keys in scripts
- Audit every change to retention and access

Coordinating ERP and identity platform recovery

Enterprise applications span database, application, and integration tiers. Use application-consistent backups and documented recovery order.

1

Database

Recover first.

2

Application tiers

Bring up next.

3

Interfaces

Reconnect last.

Recovery plans from Session 4 can sequence VM tiers.

Demonstration: RMAN to an immutable target, with a restore to prove it

1. Run an RMAN backup from Oracle on an Azure VM to a Blob container with time-based retention
2. Attempt to tamper with the backup set
3. Perform a point-in-time restore to a second instance
4. Validate the synthetic ERP dataset in the restored database

A backup that cannot be altered, and a restore that proves it was good, in one flow.



Break - 15 minutes

Back for other database platforms and restore validation.

BLOCK 3 · 40 MIN

Other database platforms and restore validation

SQL Server, Azure SQL, MySQL, PostgreSQL, and the proof that restores work.



SQL Server in Azure VMs: Azure Backup workload protection



Vault-based

Azure Backup protects SQL Server in Azure VMs from a Recovery Services vault.



Point in time

Full, differential, and log backups with point-in-time restore.



Inherited controls

Immutable vault, soft delete, and multi-user authorization from Session 3.

Azure SQL: automatic backups and long-term retention

1–35

Days of short-term retention

Point-in-time restore within that window.

10 yrs

Long-term retention

Up to 10 years, stored in redundant Blob storage.

~10 min

Log backup interval

Full weekly, differentials every 12 or 24 hours. Hyperscale uses storage snapshots for fast backup and restore.

Azure Database for MySQL and PostgreSQL



Automatic backups with point-in-time restore within the configured retention period



Geo-redundant backup storage enables geo-restore to another region



Azure Backup vaulted backup for PostgreSQL flexible server is generally available: long-term retention up to 10 years in a Backup vault with immutability



Azure Backup long-term retention for MySQL flexible server is a paused preview; use native backups and export for long-term copies today



Confirm options per flexible server configuration in Microsoft Learn

Platform protection at a glance

Platform	Primary protection	Long-term or immutable copy
Oracle on Azure VM	RMAN plus Azure Backup	Immutable Blob container
Oracle AI Database@Azure	Oracle-managed backup, including Zero Data Loss Autonomous Recovery Service	Azure targets per service options
SQL Server in Azure VM	Azure Backup workload backup	Immutable Recovery Services vault
Azure SQL	Automatic backups, 1-35 days	Long-term retention up to 10 years
MySQL / PostgreSQL	Automatic backups, geo-restore	Backup vault for PostgreSQL flexible server (up to 10 years); MySQL long-term retention via native export preview is paused

Restore validation: proof, not hope

01

Automate

Restore to an isolated environment on a schedule.

02

Check integrity

Run integrity checks (for example RMAN VALIDATE) and application queries.

03

Measure

Compare actual restore time against RTO.

04

Record

Store results as audit and leadership evidence.

Recovery rehearsal cadence

Critical databases

Frequent automated restore tests plus a full rehearsal on a set schedule.

Important databases

Regular sampled restores.

Every rehearsal

Updates the runbook and the measured RTO.

BLOCK 4 · 75 MIN

Design exercise and leadership pitch review

Turn today's architecture into a design and a pitch leadership can act on.



Design exercise: the immutability tier for the enterprise Oracle estate

01	02	03	04	05
Backup targets	Immutability policy and retention	Separation of duties	Restore testing cadence	Estimated monthly cost

Plus three sentences a vice president could use to explain why the business should care.



Translating immutability into risk language

From "WORM container" to **"a backup no attacker or insider can erase"**

From "restore test" to **"proven ability to run payroll after an attack"**

- Quantify protection cost per month against the cost of an outage of ERP or identity systems
- Present two or three options with clear trade-offs

How the 75 minutes run

1

Design

Groups design the target-state architecture.

2

Draft

Draft the three-sentence leadership pitch.

3

Present

Present design and pitch.

4

Review

Review against Microsoft and Oracle reference guidance.

Key takeaways and what comes next

Takeaways

- Choose the Oracle path by operating model, performance, and licensing
- RMAN to immutable Blob, aligned retention, and separation of duties form the immutability tier
- Restore validation turns protection into evidence

Materials

Demo script, Enterprise Application Dataset, Oracle on Azure decision guide, business case template

Next

FinOps, data management as code, and the roadmap capstone

Capstone input: Immutability tier for the Oracle estate